



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2028	M.TECH Dual Degree 5Y	IIT Kharagpur	9.17 / 10
2023	AISSCE(CBSE XII)	Ekalavya Educational Complex	88.8%
2021	AISSE(CBSE X)	Delhi Public School	95.4%

INTERNSHIPS

Computer Vision Research Intern Department of Civil Engineering IIT Roorkee	May 2026 - Jul 2026
Objective: Developing AI-driven perception and analytics systems for intelligent transportation applications Guide: Prof. Indrajit Ghosh	
- Evaluated AutoSpeed, EgoLanes models for CIPO detection, lane perception and vehicle interaction analysis on monocular Indian road videos - Performed camera calibration using 36 checkerboard frames (0.45 px RMS error) and built vanishing point estimation, and metric BEV pipelines - Built a web-based traffic analytics platform for camera calibration, speed estimation, and metric measurement from monocular videos	
Strategy Intern Tata Motors Pune, India	Jun 2025 - Aug 2025
Objective: Conducted technology assessment , market intelligence , and competitive analysis to support commercial vehicle strategy decisions	
- Evaluated government policies, incentives, and regulatory frameworks to assess market impact & support strategic business decisions - Performed competitive benchmarking across global OEMs, evaluating EV, hydrogen, ADAS, SDV & V2X technology roadmaps comprehensively - Developed strategic recommendations for electrification and connected mobility using market research, policy insights & technology trends	

PROJECTS

Explainable Credit Risk Intelligence Platform Self Project	May 2026
- Built a credit risk prediction platform on 150K+ loan records using XGBoost, achieving ROC-AUC 0.86, Recall 76.5% for default detection - Benchmarked 4 ML models using SMOTE, class weighting, and feature engineering, improving default prediction on imbalanced credit data - Integrated SHAP explainability & an AI Credit Assistant, deploying the solution via FastAPI/Next.js for transparent lending recommendations	
Streamflow Prediction Course Project Prof. Rajib Maity	Apr 2026
- Developed daily streamflow prediction models using rainfall and lagged discharge features with ML methods (Random Forest, LightGBM) - Implemented sequence and modern time-series architectures (LSTM, Transformer models) to capture underlying temporal flood dynamics - Used SHAP-based Explainable AI to interpret model predictions and identify key rainfall and discharge factors driving high-flow & flood risk	
Image Classifier Deep Learning Self Project	Jun 2025
- Developed and trained a TensorFlow CNN architecture on 70,000+ grayscale Fashion-MNIST images for multi-class apparel classification Optimized model performance via normalization, hyperparameter tuning, and validation monitoring, achieving 88% test accuracy - Conducted detailed class-wise error analysis using precision, recall, and F1-scores to fully understand model misclassification patterns	
News Category Classifier NLP Self Project	May 2025
- Developed an end-to-end text classification pipeline categorizing 2,200+ BBC articles into 5 classes using TF-IDF and Logistic Regression - Engineered textual features through NLTK preprocessing, lemmatization, stopword filtering, and vectorization for multi-class classification - Delivered 96% test accuracy and integrated a real-time prediction interface enabling instant classification of user-provided news articles	

COMPETITION/CONFERENCE

General Championship, Technology Students' Gymkhana IIT Kharagpur	Mar 2025 - Apr 2025
- Derived 30+ features from weather, holidays, discount and analyzed product-channel links using correlation and performance scores - Built interactive Power BI dashboards to visualize trends, category-level insights, and actionable metrics for business stakeholders - Collaborated in a 20-member analytics team to synthesize actionable insights and support data-driven marketing strategy decisions	

SKILLS AND EXPERTISE

Languages: C, Python, SQL, HTML, CSS, JavaScript	Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras, PyTorch
Tools: Jupyter Notebook, Visual Studio Code, Google Colab, MS Excel, MS PowerPoint, Canva, Power BI, Tableau, Streamlit, GitHub	

COURSEWORK INFORMATION

Maths: Advanced Calculus Linear Algebra Numerical and Complex Analysis Probability and Statistics Core: Solid Mechanics Structural Analysis Design of RC & Steel Structures Structural Modelling Lab Transportation Engineering Others: Programming & Data Structures AI: Foundation & Applications Time Series Analysis* Deep Learning* (*online courses)

POSITIONS OF RESPONSIBILITY

Design Team Head Megalith - Civil Engineering Tech Fest IIT Kharagpur	Jun 2024 - Present
- Led design and outreach initiatives for Megalith 26, creating promotional content that reached 5,000+ students and professionals - Coordinated on-ground execution for 15+ events, ensuring smooth operations, seamless logistics, and participant engagement	

EXTRA CURRICULAR ACTIVITIES

National Cadet Corps (NCC) Army Wing Cadet	Jul 2023 - Jul 2025
Served as an NCC Cadet for two years, earning the B - Certificate and recognition as Best Cadet at Combined Annual Training Camp	